Semi-Supervised Neural Architecture Search for Hyperspectral Imagery Classification Method With Dynamic Feature Clustering

Wei Wei, *Senior Member, IEEE*, Shuyi Zhao, Songzheng Xu, Lei Zhang, *Member, IEEE*, and Yanning Zhang, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) contains rich spatial and spectral information, which makes the HSI classification task the research focus of HSI analysis within remote sensing community. Though deep learning-based HSI classification methods obtain good performance in recent years, how to learn network structure better suitable for a given HSI instead of utilizing a manually designed one for HSI classification is still a challenging problem, especially providing only a small amount of labeled samples. To address this problem, we propose the first semi-supervised HSI classification network constructed via the neural architecture search (NAS). Specifically, we propose a two-head semi-supervised HSI classification framework utilizing both labeled and unlabeled data, which consists of a shared feature extraction module, a classifier module for labeled samples together with a clustering module for unlabeled samples. To boost the performance of the constructed two-head network, we propose to utilize deep features instead of the original pixels for HSI clustering to generate pseudo-labels for the unlabeled data. Within the conducted semi-supervised network, we specifically design a method to automatically search for the shared feature extraction module better suitable for the given HSI data, which leads to better HSI classification results. Experimental results on three HSI datasets demonstrate the effectiveness of the proposed method, providing only a limited number of labeled training samples.**

***Index Terms*— Hyperspectral image (HSI) classification, neural architecture search (NAS), semi-supervised learning.**

I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D image data capturing both spectral information and spatial information. Compared with conventional RGB images, the continuous spectral data of each pixel [1], [2], [3] enable HSI powerful ability to identify different materials. As a result, HSI has been widely utilized in a lot of fields, including ecological science [4], geological science [5], and agriculture [6].

Within the HSI community, HSI classification is an essential and fundamental task, which aims to assign a specific category label to each pixel within HSI. For this purpose, different kinds of HSI classification methods have been proposed to date, which can be generally divided into handcrafted feature-based methods [7], [8], [9] and deep learning-based ones [10], [11], [12], [13], [14], [15], [16], [17], [18], [19]. Handcrafted feature-based method consists of two independent parts, including a feature extraction module and a classifier module. The feature extraction module is used to extract the features of HSI and the classifier module is used to categorize the pixels based on the extracted features. However, the feature extraction module can only extract shallow features with limited representation ability and is independent of the classifier, which limits the classification performance of the handcrafted feature-based methods.

In recent years, with the development of deep learning techniques, deep learning-based methods have been introduced into HSI classification and achieved great progress [10], [11], [12], [13], [14], [15], [16], [17], [18], [19]. Compared with the handcrafted feature-based HSI classification method, the deep feature-based method can effectively extract the deep feature of HSI. Besides, it can obtain more discriminative features by intergrating feature extraction and classifier into one framework, which is conducive to enhance the classification performance. Nevertheless, the following two limitations impede further improvement of the current deep learning-based method.

1) Deep learning-based methods always demand large amount of labeled data for training, which, however, is hard to be satisfied since labeling HSI is difficult in reality [20], [21], [22]. Although the semi-supervised method provides a way to address this problem, most of the current semi-supervised methods focus on the original pixel of the images, ignoring deep features that have better semantic distribution characteristics.
2) The network structures are manually designed, which may not accurately reflect the characteristics of a given HSI data. Despite the fact that there have been some attempts to find data-specific structure for HSI

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Wei Wei is with the Shaanxi Provincial Key Laboratory of Speech and Image Information Processing and the National Engineering Laboratory for Integrated Aerospace-Ground-Ocean Big Data Application Technology, School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China, and also with the Research and Development Institute, Northwestern Polytechnical University, Shenzhen 518031, China (e-mail: weiweinwpu@nwpu.edu.cn).

Shuyi Zhao, Songzheng Xu, Lei Zhang, and Yanning Zhang are with the Shaanxi Provincial Key Laboratory of Speech and Image Information Processing and the National Engineering Laboratory for Integrated Aerospace-Ground-Ocean Big Data Application Technology, School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: zhao_sy@mail.nwpu.edu.cn; songzhengxu@mail.nwpu.edu.cn; zhanglei211@mail.nwpu.edu.cn; ynzhang@nwpu.edu.cn).

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classification, the existing methods are fully supervised, which rely heavily on a large number of labeled samples.

To further improve the performance of HSI classification providing only small amount of labeled training samples, we propose a semi-supervised structure-adaptive HSI classification method in this study, which can automatically construct network structure better suitable for the given HSI data. Specifically, we construct a two-branch network with a shared feature extraction module, a classification module for labeled samples, along with a clustering module for unlabeled samples, from which large amount of unlabeled samples can be introduced into HSI classification training. Within the constructed network, we parameterize and continuousize the discrete network structure of the shared feature extractor, and utilize a gradient-based optimization method to automatically generate the shared feature extractor, which dominates the majority parameters of the network. In addition, we perform clustering based on deep features rather than the original pixels of the unlabeled samples, accompanied with a dynamic memory module to alleviate sample feature changing caused by the update process of network parameters. Experimental results on three HSI datasets demonstrate the effectiveness of the proposed method.

In summary, the main contributions of this study can be summarized as follows.

1) We explore ways to construct semi-supervised HSI classification network via neural architecture search (NAS). To the best of our knowledge, this is the first work attempts to construct a semi-supervised HSI classification network automatically.
2) We propose a two-head semi-supervised HSI classification framework utilizing both labeled and unlabeled data. Within the conducted semi-supervised HSI classification framework, we design a method to automatically search for the shared feature extraction module better suitable for the given HSI data.
3) We utilize the dynamic feature clustering method together with loss reweighting and reclustering to generate the pseudo-labels better suitable for the unlabeled data, which also improves the performance of the proposed semi-supervised HSI classification method.

II. Related Works

A. Deep Learning-Based HSI Classification

With the development of deep learning, deep learning-based HSI classification methods have been extensively studied. Chen et al. [10] first introduced deep learning for HSI classification, which utilizes stacked autoencoders (SAEs) to extract high-level features from HSI. Ma et al. [11] considered sample similarity and added it as a regularization item for updating features. A deep belief network-based HSI classification method is proposed by Chen et al. [14], which consists of principal component analysis, deep belief network, as well as logistic regression method. Hu et al. [15] exploited a five-layer convolutional neural network (CNN) directly utilizing spectral information for HSI classification. Yue et al. [16] proposed a deep CNN (DCNN) utilizing both spectral and spatial information within HSI, combined with the logistic regression method. Ying et al. [17] proposed a 3-D CNN to accomplish the HSI classification, where 3-D convolution can be adopted to extract the deep spectral–spatial features.

Besides, other kinds of network architectures are also utilized for HSI classification. Ying et al. [17] proposed a spatial–spectral residual network (SSRN) for HSI classification. Within this network, residual blocks are utilized to facilitate the gradient backpropagation process, and batch normalization is imposed to improve the classification accuracy. Paoletti et al. [19] introduced dense connection into the hyperspectral classification network, which can extract spectral–spatial features effectively and enhance the generalization ability of the proposed network.

Although deep learning-based HSI classification methods have made remarkable progress, they often suffer from the following two shortcomings. First, most deep learning-based methods require a large number of labeled data for training. Unfortunately, the annotation process of HSI requires the experience of experts and the available labeled data are often limited, which leads to the overfitting problem and degenerates the classification performance. Second, the architecture of deep neural network is usually manually designed by experts, which is usually not the optimal structure for a specific dataset. As a result, the performance of HSI classification will also be affected. In order to solve the problems above, we propose a semi-supervised HSI classification method, which can search for a suitable network structure.

B. Semi-Supervised Deep Learning in HSI Classification

The purpose of semi-supervised learning is to learn the data distribution not only from a limited amount of labeled data but also from a large amount of unlabeled data.

To generate more labeled training data from existing labeled training data for semi-supervised learning, some studies focus on semi-supervised methods based on generative adversarial networks (GANs) [23], [24], [25]. Liang et al. [24] proposed a semi-supervised spectral–spatial attention feature extraction approach based on the GAN (SSAT-GAN), in which unlabeled data are utilized to enhance the reconstruction ability of the generator and spectral–spatial attention modules are implemented to extract discriminative features. Li et al. [25] proposed a variational GAN with crossed spatial and spectral interactions (CSSVGAN), in which a dual-branch variational encoder is adopted to extract spectral–spatial features. Within CSSVGAN, a crossed interactive generator is utilized to generate virtual samples, and a discriminator combined with classifier is exploited for accurate classification. However, the above methods rely heavily on the existing labeled data. Once the labeled training data are severely limited (i.e., only few labeled training data are provided), the performance of these methods will degenerate.

Some attempts turn to extend labeled training set by assigning pseudo-labels to unlabeled data. Wu and Prasad [26] proposed to pretrain the deep network with unlabeled data and their pseudo-labels first and then fine-tune the network with limited labeled data. Kang et al. [27] proposed a

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deep feature fusion network and generated pseudo-labels by a decision function based on classification probability. In addition to pseudo-label-based semi-supervised learning methods, the structure information of unlabeled data can also be utilized to regularize classification models. Fang et al. [28] proposed a semi-supervised residual neural network with a cotraining strategy, which guides the network to learn the information from unlabeled data. Zhou et al. [29] proposed a semi-supervised classification method with cotraining autoencoder. Wei et al. [30] proposed a deep cube-pair network (DCPN) for HSI classification, trying to exploit the similarity between pairs of samples to generate more samples for training, so as to regularize the network training process with unlabeled samples. Jia et al. [31] proposed a semi-supervised Siamese network that embeds the Siamese network into a semi-supervised learning scheme.

In addition, some methods focus on semi-supervised HSI classification based on graph convolutional networks (GCNs) [32], [33], [34]. Sha et al. [32] proposed a novel semi-supervised network based on graph attention network (GAT). The method designs a spatial–spectral joint measurement to utilize the spatial information of HSI and perform graph convolution based on the attention coefficients of different neighboring nodes. Wang et al. [33] utilized graph convolution operation to extract discriminative spectral–spatial features from HSI, and a broad learning system (BLS) is applied to perform broad expansion on features, which enhances the representation ability of the network. Xi et al. [34] proposed a cross-scale graph prototypical network (XGPN) to capture the neighborhood information of different scales, which further improves the performance of semi-supervised HSI classification.

Although the above semi-supervised-based HSI classification methods perform well, the network architecture is typically designed manually. They cannot accurately reflect the characteristics of the given HSI data, as was already discussed, which limits the classification performance of the network. In view of the above problem, we propose a semi-supervised HSI classification method based on a neural architecture search algorithm, which can design a suitable network structure automatically for given HSI data.

C. Neural Architecture Search

Different from the traditional handcrafted deep learning algorithms, the neural architecture search algorithm aims to design network architecture automatically, which can find the optimal network structure for the specific data. According to the search strategy adopted, most NAS algorithms can be divided into three categories, including evolutionary algorithm (EA)-based NAS [35], [36], [37], reinforcement learning (RL)-based NAS [38], [39], [40], and gradient-based NAS [41], [42].

A few efforts have been made to search data-specific structures for HSI classification. Inspired by NAS, Chen et al. [43] proposed Auto-CNN to automatically design the HSI classification network, which first compresses the spectral dimension of HSIs before searching like DARTS [41]. However, Auto-CNN [43] is trained in a supervised manner, thus relying on a large amount of labeled data. Different from the above supervised method, the proposed method is semi-supervised, which can make full use of the information in a large amount of unlabeled data.

III. Proposed Method

In this study, we denote labeled and unlabeled hyperspectral datasets as $D_L = \{x_i, y_i\}_{i=1}^{M}$ and $D_U = \{x_i\}_{i=M+1}^{M+N}$, where $x_i \in \mathbb{R}^{d\times 1}$ represents a pixel with a d-dimensional spectrum and $y_i \in [1, 2, \ldots, K]$ represents its corresponding label. M and N, respectively, represent the number of labeled samples and unlabeled samples in the training set. K is the total number of classes in the dataset. The goal of HSI classification is to establish a mapping function between hyperspectral data x_i and its labels y_i.

Deep learning-based HSI classification methods usually require a large amount of labeled data for training. However, since pixel-level labeling of HSIs is difficult and expensive, only a few labeled samples can be provided for training, which can easily lead to network overfitting and lower classification performance. To address this issue, some researchers have proposed semi-supervised learning-based methods for HSI classification [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]. However, the current semi-supervised methods focus on the original pixel of samples, while the deep features of samples have better semantic distribution characteristics.

In addition, most of the existing deep learning-based network structures are subjectively manually designed by researchers. However, due to the characteristics of the hyperspectral imaging system, the HSIs collected by different sensors have different spectral bands, spectral ranges, and spatial resolutions [44]. Therefore, the network structure that can provide the best classification performance for various HSIs also varies. As a result, the network structures that are subjectively manually designed by researchers may not necessarily achieve the best classification performance.

In this study, we try to address the problem from the following two perspectives. On the one hand, we propose a semi-supervised HSI classification method based on dynamic feature clustering, which clusters the deep features of unlabeled samples instead of original pixels to improve the performance of HSI classification in the case of few samples. On the other hand, the network structure of the feature extraction module is automatically designed by the neural architecture search algorithm, which makes the model data-adaptive and further improves the classification performance.

In the following, we will first briefly introduce the overall framework of the proposed method in Section III-A. Then, the two main works of this work are introduced in detail, including dynamic feature clustering in Section III-B and network structure search algorithm in Section III-C. Finally, we summarize the training process of the algorithm in Section III-C2.

A. Overall Framework

The overall framework of the proposed method is shown in Fig. 1. The algorithm consists of two main components: 1) a dual-branch network containing a shared feature extraction module and a classification module together with a clustering

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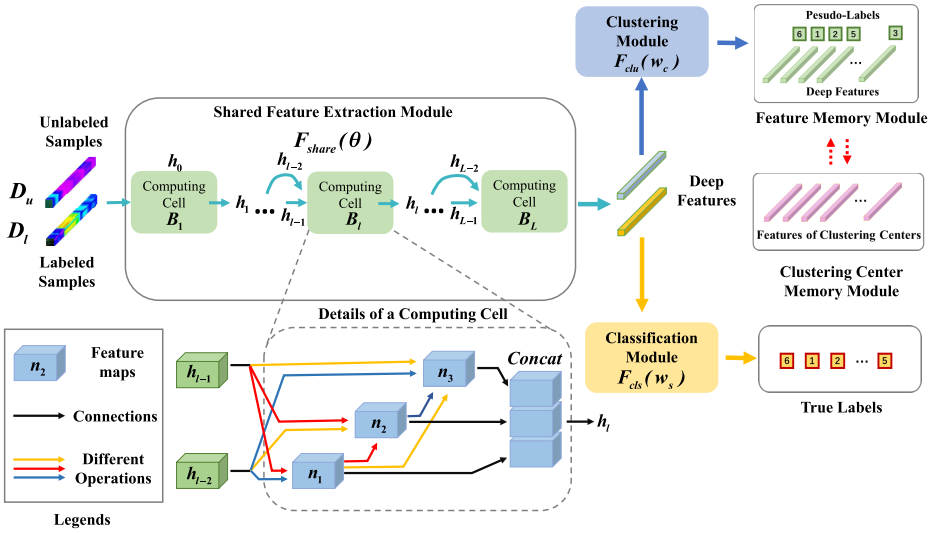


Fig. 1. Overall framework of the proposed method. A dual-branch network and two memory modules are utilized to perform dynamic deep feature-based semi-supervised learning. Several computing cells are stacked to form the shared feature extraction module, and a gradient-based architecture search strategy is implemented to design the optimal architecture automatically.

module and 2) dynamic memory modules including a feature memory module and a clustering center memory module.

Specifically, within the dual-branch network, the shared feature extraction module $F_{\text{share}}(\theta)$ consisting of several computing cells B_i is used to extract deep features of the HSI. Concretely, the computing cell B_i can be denoted as a directed acyclic graph $B(\mathbf{N}, \mathbf{E})$, where $\mathbf{N}$ is the collection of nodes $n^{(i)}$ representing the feature graph in the network and $\mathbf{E}$ is the collection of edge $e(i, j)$ indicating the operation to transform the ith node into jth node. The classification module $F_{\text{cls}}(w_s)$ classifies the labeled data, while the clustering module $F_{\text{clu}}(w_c)$ clusters the unlabeled data with pseudo-labels produced by the dynamic memory module. The dynamic memory modules include a feature memory module to store the deep features and clustering pseudo-labels of the samples, and a clustering centers memory module to store clustering centers. The deep features and clustering pseudo-labels stored in memory modules will be smoothly updated during the training process to alleviate sample feature altering brought on by the updating of network parameters.

It is worth noting that the proposed semi-supervised neural architecture search method is dedicated to search the optimal structure of the shared feature extraction module $F_{\text{share}}(\theta)$. Similar to earlier research [41], [42], we search for a computing cell with the optimal structure instead of the entire network to lower the computational cost, and then, the cells will be stacked to form the final shared feature extraction module. Within the constructed network, we utilize the gradient-based optimization process to automatically generate the shared feature extractor. Specifically, we parameterize and continuousize the discrete network structure of the shared feature extractor and then optimize the structure parameters α^* and network parameters θ^* by a bilayer optimization strategy [45].

B. Dynamic Feature Clustering

The current semi-supervised methods mainly focus on the original pixels of samples [20], [21], [22], while the deep features have better semantic distribution characteristics. In order to improve the classification performance of deep neural network with limited labeled samples, we construct a dynamic feature clustering-based semi-supervised training scheme. The proposed method generates pseudo-labels for unlabeled data. The process of dynamic feature clustering is shown in Fig. 2. During the optimizing process of network parameters, the implemented semi-supervised training scheme consists of the following four steps.

Step 1: The labeled and unlabeled samples x are input into a shared feature extraction module to get their features $f = F_{\text{share}}(\theta, x)$. Then, the features f are sent into the classification module F_{cls} to get the prediction labels $\hat{y}$ or the clustering module F_{clu} to get the cluster label $\hat{z}$.

Step 2: The total loss of the dual-branch network L_{total} consists of two parts: the classification loss of the supervised branch L_{cls} and the feature clustering loss of the unsupervised branch L_{clu}. L_{total} can be calculated as follows:

$$\begin{aligned} L_{\text{total}} &= L_{\text{cls}} + \lambda L_{\text{clu}} \\ &= -\left(\sum y \log(\hat{y}) + \lambda \sum z \log(\hat{z})\right) \end{aligned} \tag{1}$$

where y is the true label of the labeled sample and z is the pseudo-label of the unlabeled sample obtained from the feature memory module. λ represents the weight of the feature clustering task loss. Then, the network parameters can be optimized by minimizing the total loss with backpropagation algorithm.

Step 3: The features f_m in the feature memory module are updated with the deep feature f of the unlabeled samples, which are extracted by the shared feature extraction module as shown in (2). Pseudo-labels z of the modified features are then computed based on their distances from various cluster centers. As shown in (3), the pseudo-class whose feature has the smallest distance to different cluster centers is then

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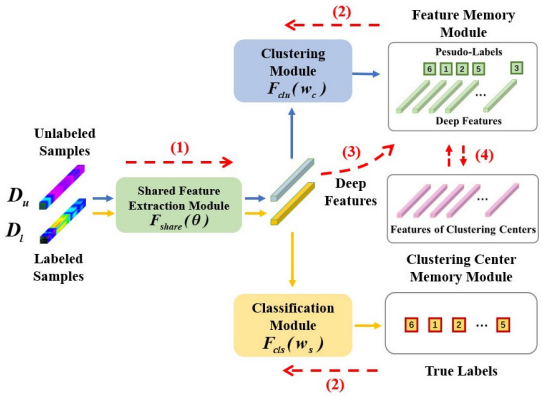


Fig. 2. Semi-supervised training process based on dynamic feature clustering consists of four steps: (1) input labeled and unlabeled samples into network to obtain their predicted labels, (2) obtain pseudo-labels from feature memory module and optimize network parameters by minimizing the total loss L_{total}; (3) update deep features and clustering pseudo-labels in the feature memory module with newly computed deep features, and (4) update clustering center memory module.

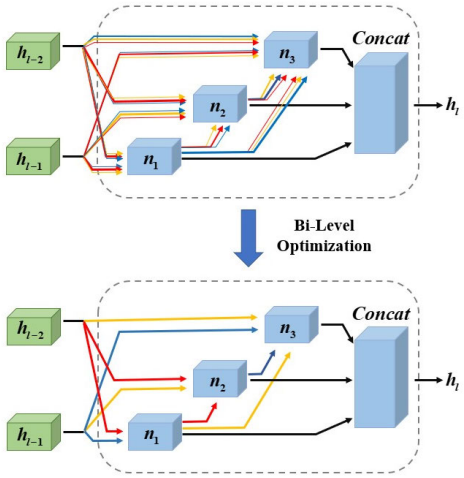


Fig. 3. Gradient-based neural architecture search algorithm. The algorithm can relax a discrete operation selection process to a weighted sum of mixed operations and then optimize the architecture parameters on validation set. Finally, the operation with the highest weight is retained to form the final architecture.

assigned as the new pseudo-label

$$f_m = \beta \frac{f}{||f||_2} + (1-\beta) f_m \tag{2}$$

$$z = \underset{z\in\{1,2,\ldots,C\}}{\arg\min} ||f - C_z||_2 \tag{3}$$

where β is the update parameter that attempts to perform soft updates to stabilize the training process and C_z represents the feature of the cluster center of class z.

Step 4: The clustering centers C_z in the clustering center memory module are updated every k iterations according to the average of the features of all unlabeled samples belonging to the corresponding pseudo-class.

It is worth noting that when dynamic feature clustering is performed, there may be class imbalance, causing the feature clustering task overpay attention to some pseudo-classes with a large number of samples and ignore the pseudo-classes with a small number of samples. As a result, the performance of the network will be influenced. To alleviate the problem, the following two improvements are made in this study as follows to improve the performance.

1) Loss Reweighting: To balance the contributions from each class during training, different weights are assigned to their corresponding losses according to the amount of samples within the pseudo-class. Specifically, when calculating the loss of the clustering branch, the corresponding weight w is positively correlated with $(1/(N_C)^{1/2})$, where N_C represents the quantity of samples in the sample feature memory module that belonging to class C. As a result, the feature clustering loss is calculated with

$$L_{\text{clu}} = -\sum_{i=1}^{K} w_i z_i \log(\hat{z}_i) \quad \text{s.t.} \quad w_i \propto \frac{1}{\sqrt{N_C}}. \tag{4}$$

2) Reclustering: Pseudo-classes with extreme numbers of samples will be reclustered. Specifically, for the pseudo-class with too many samples, it will be reclustered into two subclasses by the K-means algorithm. A pseudo-class with few samples will be combined with the one that is closest to it.

C. Neural Architecture Search

The neural architecture of most current semi-supervised-based HSI classification methods [26], [27], [28], [29], [30] is manually designed by experts. However, in practical applications, it is difficult for manually designed network structures to achieve the best results for the specific dataset.

In order to solve the above problems, we adopt the neural architecture search algorithm to automatically design the network structure. Specifically, within the proposed HSI classification method, the feature extraction module plays a crucial role, as the HSI features need to be well representative and discriminative for more accurate clustering and assignment of pseudo-labels. Therefore, in order to automatically design the optimal feature extraction module for various situations, a gradient-based neural architecture search algorithm is designed as follows.

1) Search Space: Directly searching the overall structure of the deep network will result in a very broad search space, a protracted search time, and a considerable computational cost. Therefore, inspired by the recent progress of neural architecture search method [41], [42], the proposed HSI classification method searches for a computing cell B before stacking L computing cells to form the final shared feature extraction module within the dual-branch network. The computing cell's structure can be described as a directed acyclic graph, as shown in Fig. 3. Each node $n^{(i)}$ represents a feature map in the network, and each edge $e(i, j)$ in the graph represents some operations $o(i, j)$ that transforms the ith node to the jth node (e.g., convolution, pooling, and skip connection). Following the previous study [41], [46], the computing cell B_l contains one output node and two input nodes h_{l-1} and h_{l-2}, which are the outputs of two predecessor computing cells B_{l-1} and B_{l-2}. The output of a computing cell h_l is generated by concatenating all nodes

$$h_l = \text{Concat}(n_{(i)} | i \in \{1, 2, \ldots, N\}) \tag{5}$$

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TABLE I
CANDIDATE OPERATIONS

Operation Name	Details of Operations
conv_3-3	ReLU-Conv(1, 3×3)-BN
conv_5-5	ReLU-Conv(1, 5×5)-BN
conv_7-7	ReLU-Conv(1, 7×7)-BN
SepConv_3-3	ReLU-SepConv(1, 3×3)-BN
SepConv_5-5	ReLU-SepConv(1, 5×5)-BN
SepConv_7-7	ReLU-SepConv(1, 7×7)-BN
max_pool_3-3	MaxPooling(3×3)
avg_pool_3-3	AveragePooling(3×3)
skip-connection	$f(x) = x$
zero-option	$f(x) = 0$

where the value of each node $n(i)$ is computed based on the values of all predecessor nodes

$$n^{(i)} = \sum_{j<i} o^{(j,i)}(n^{(j)}). \tag{6}$$

For each computing cell, the stride of the last convolution operation is set to 2, and the number of channels is doubled to capture the deep features. At the end of all stacked computing cells, a global average pooling is implemented to obtain the final 1-D feature. As shown in Table I, the set of candidate operations O consists of ten operations, each of which represents a function $o(\cdot)$ to be applied to node $n^{(i)}$.

2) Gradient-Based Search Strategy: The neural architecture search algorithm essentially selects the optimal operation o^* for each edge $e^{(i,j)}$ from all candidate operations O in a computing cell. In order to optimize the architecture by a gradient-based method, the discrete search space needs to be converted into a continuous space, and the objective function becomes a differentiable function [41]. We relax the discrete operation selection process to softmax sum over all possible operations as follows:

$$\overline{o}^{(i,j)}(n) = \sum_{o\in O} \frac{\exp(\alpha_o^{(i,j)})}{\sum_{o'\in O}\exp(\alpha_{o'}^{(i,j)})} o(n) \tag{7}$$

where $\alpha_o^{(i,j)}$ denotes the weight of an operation $o^{(i,j)}$. In fact, the output of mixed operations $\overline{o}^{(i,j)}$ is the weighted sum of each operation in candidate operations O. Then, the purpose of neural architecture search is simplified to learn a set of parameters $\alpha = \{\alpha^{(i,j)}\}$, which we will refer to as architecture parameters. At the end of the search algorithm, the most likely operation $o^{(i,j)*} = \arg\max_{o\in O}\alpha_o{}^{(i,j)}$ will be selected as the ideal operation in the final network. In addition, we refer to the parameters in a specific operation as θ, i.e., the weights of the convolution filter.

After relaxation, how to find the best architecture parameters α^* and network parameters θ^* is critical to the final performance. For this purpose, we adopt a bilevel optimization method [45] to co-optimize θ and α. Specifically, we optimize the network parameters θ by minimizing the training loss L_{train} ($L_{\text{train}} = L_{\text{total}}$) and optimize the architecture parameters α by minimizing the validation loss L_{val} ($L_{\text{val}} = L_{\text{cls}}$). Thus, the gradient-based optimization strategy based on bilevel optimization can be described as follows:

$$\begin{aligned} &\min_{\alpha}\{L_{\text{val}}(\theta^*(\alpha), \alpha)\} \\ &\text{s.t. } \{\theta^*(\alpha) = \arg\min_{\theta} L_{\text{train}}(\theta, \alpha)\}. \end{aligned} \tag{8}$$

After the optimization process, the operation with the highest weight coefficients c is retained for the final architecture.

Algorithm 1 Dynamic Feature Clustering-Based Semi-Supervised Neural Architecture Search

Require: Labeled data D_L, Unlabeled data D_U, initial training epochs T_0, total training epochs T, refresh frequency of memory module.

Initialize computing cells randomly, and stack them to form the dual-branch network.
Cluster the original unlabeled samples to get their pseudo-labels $\hat{z}$.
while $t \leq T_0$ **do**
  Input labeled and unlabeled data x into network to obtain their predicted classification label $\hat{y}$ and clustering label $\hat{z}$.
  Optimize network parameters by minimizing the total loss L_{total}.
end while
Extract deep features f of samples x with F_{share}.
Cluster the deep features f to get their pseudo-labels $\hat{z}$, and initialize both memory modules.
while $t \leq T$ **do**
  Fix the architecture parameters α.
  Input labeled and unlabeled samples into dual-network to get their predicted label $\hat{y}$ and $\hat{z}$.
  Obtain feature clustering pseudo-label z from sample feature memory module, and optimize the network parameters θ according to equations 1.
  Update sample feature memory module with newly extracted deep features according to equations 2 to 3.
  if $t\%k == 0$ **then**
    Refresh clustering center memory module.
  end if
  Fix the network parameters θ, and optimize architecture parameters by minimizing validation loss L_{val}.

$$\min_{\alpha}\{L_{val}(\theta^*(\alpha), \alpha)\}$$

end while
Retain operations with the highest weight coefficient to form the final architecture.

D. Training Strategy

In this section, we will describe in detail the training process of the dynamic feature clustering-based semi-supervised neural architecture search algorithm. Specifically, the proposed semi-supervised neural architecture search algorithm consists of two phases: initialization phase and neural architecture search phase.

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TABLE II
DETAILS OF DIFFERENT DATASET

No.	Indian Pines		University of Pavia		Houston 2013	
	Category	Samples	Category	Samples	Category	Samples
1	Corn-notill	1428	Asphalt	6631	Healty Grass	1251
2	Corn-mintill	830	Meadows	18649	Stressed Grass	1254
3	Grass-pasture	483	Gravel	2099	Synthetic Grass	697
4	Grass-tree	730	Tree	3064	Trees	1244
5	Hay-windrowed	478	Sheets	1345	Soil	1242
6	Soybean-notill	972	Bare Soil	5029	Water	325
7	Soybean-mintill	2455	Bitumen	1330	Residential	1268
8	Soybean-clean	593	Bricks	3682	Commercial	1244
9	Woods	1265	Shadows	947	Road	1252
10					Highway	1227
11					Railway	1235
12					Parking Lot 1	1233
13					Parking Lot 2	469
14					Tennis Court	428
15					Running Track	660

1) Initialization Phase: First, the randomly initialized computing cells B will be stacked to form the shared feature extraction module in the dual-branch network, and the original unlabeled samples will be clustered to obtain their pseudo-labels z. Second, both labeled and unlabeled data will be used to jointly train the two-branch network. The initialized shared feature extraction module then extracts the deep features $f = F_{\text{share}}(\theta, x)$ of unlabeled samples, which are clustered to get pseudo-labels $\hat{z}$. Finally, the feature memory module will be initialized with deep features f and their pseudo-labels $\hat{z}$, and the clustering center memory module will be initialized with the clustering centers C_z.

2) Neural Architecture Search Phase: The neural structure and network parameters are optimized simultaneously based on a bilevel optimization strategy. Specifically, both labeled and unlabeled samples are fed into the dual-branch network to get prediction labels $\hat{y}$ and $\hat{z}$, respectively. Also, the total loss L_{total} can be calculated according to the true labels y of labeled data and pseudo-label z from memory modules as (1). Then, we can optimize network parameters θ on the training set and architecture parameters α on the validation set alternatively based on the bilevel optimization strategy. Specifically, we fix the architecture parameters α and optimize the network parameters θ by minimizing the total loss L_{total} on the training set. Next, the network parameters θ are fixed, and architecture parameters α are optimized based on the classification loss L_{val} on the validation set. Also, the two memory modules are also updated. The flow of the algorithm is shown in Algorithm 1.

IV. EXPERIMENTAL RESULTS

In this section, we conduct experiments on three commonly used public datasets to validate the effectiveness of the proposed method.

A. Hyperspectral Dataset

In this study, three commonly used hyperspectral datasets are utilized to evaluate the performance of different methods, including the Indian Pines dataset, the University of Pavia dataset, and the Houston 2013 dataset, as shown in Fig. 4. The details of different datasets are shown in Table II.

The Indian pines dataset is acquired by an Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor in Northeastern Indiana, USA. The dataset contains 145 × 145 pixels with a spatial resolution of 200 m. There are 220 spectral dimensions in the dataset, covering a wavelength from 0.4 to 2.5 μm, in which 200 spectral bands without noise are commonly utilized in experiments. The original dataset includes 16 categories, and nine categories with more samples are utilized in the experiment.

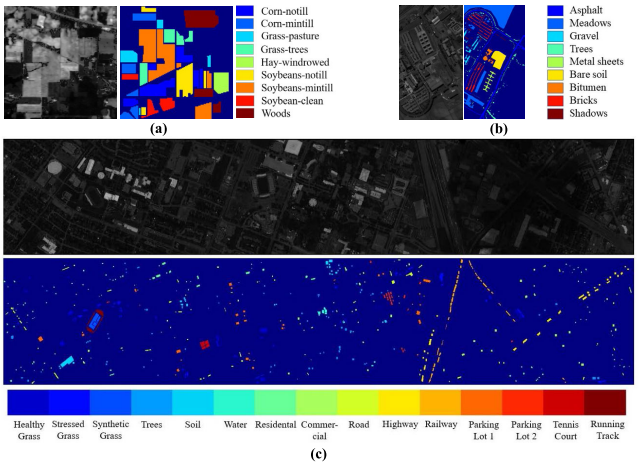


Fig. 4. Three datasets utilized in experiments. (a) 24th band and ground truth-map of Indian Pines dataset. (b) 69th band and ground-truth map of University of Pavia dataset. (c) 65th band and ground-truth map of Houston 2013 dataset.

The University of Pavia dataset is collected by Reflective the Optics System Imaging Spectrometer (ROSIS) sensor in Pavia, Northern Italy. The dataset contains 103 spectral bands ranging from 0.43 to 0.86 μm and covers 610 × 340 pixels with a geometric resolution of 1.3 m. All nine categories are utilized for experiments in this article.

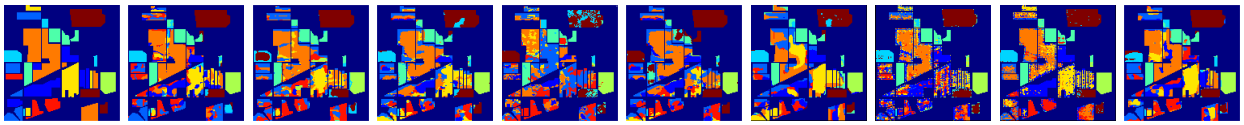
The Houston 2013 dataset is acquired by the ITRES-CASI visible near-infrared (VNIR) sensor in the University of Houston, Houston, TX, USA, which contains 349 × 1050 samples with the spatial resolution of 2.5 m. The dataset contains 144 spectral bands ranging from 360 to 1050 μm. All 15 categories are utilized in experiments.

B. Experimental Setup

The proposed method is implemented based on PyTorch and trained on a GTX1080Ti GPU. For the three datasets adopted in this study, we randomly select 20% of the samples for the test dataset. Within the remaining pixels, there are five and ten samples per class for the labeled training dataset, three and five samples per class for the validation dataset, and the remaining samples for the unlabeled training dataset. The samples are cropped with a spatial resolution of 32 × 32 and then fed into a model with a batch size of 16. The learning rate is set to 0.001 during training, and the number of computing cells is set to 4. We use the Adam optimizer to optimize architecture parameters α and the SGD optimizer to optimize network parameters θ. The final result is the average of ten random tests in order to lessen the influence of randomness on network performance.

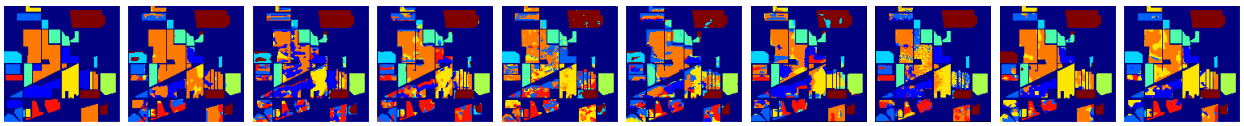
To verify the effectiveness of the proposed method, we compare the proposed method (Ours) with other deep learning-based HSI classification methods, including 3DCNN [47], pyramidal residual network (PRN) [48],

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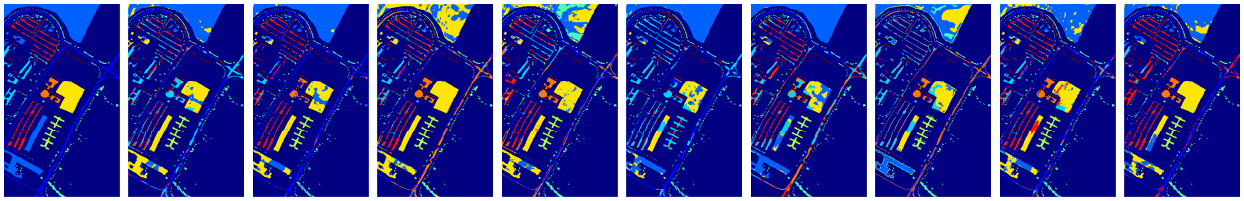
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 5. Classification maps of different methods on the Indian Pines dataset with five labeled samples per class in the training set and three labeled samples per class in the validation set. (a) Ground truth. (b) 3DCNN. (c) PRN. (d) SPCN. (e) LapSVM. (f) DCPN. (g) 3DAES. (h) XGPN. (i) AutoCNN. (j) Ours.



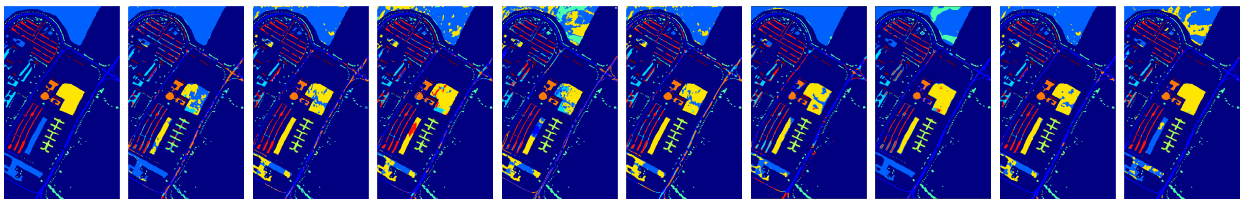
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 6. Classification maps of different methods on the Indian Pines dataset with ten labeled samples per class in the training set and five labeled samples per class in the validation set. (a) Ground truth. (b) 3DCNN. (c) PRN. (d) SPCN. (e) LapSVM. (f) DCPN. (g) 3DAES. (h) XGPN. (i) AutoCNN. (j) Ours.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 7. Classification maps of different methods on the University of Pavia dataset with five labeled samples per class in the training set and three labeled samples per class in the validation set. (a) Ground truth. (b) 3DCNN. (c) PRN. (d) SPCN. (e) LapSVM. (f) DCPN. (g) 3DAES. (h) XGPN. (i) AutoCNN. (j) Ours.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Classification maps of different methods on the University of Pavia dataset with ten labeled samples per class in the training set and five labeled samples per class in the validation set. (a) Ground truth. (b) 3DCNN. (c) PRN. (d) SPCN. (e) LapSVM. (f) DCPN. (g) 3DAES. (h) XGPN. (i) AutoCNN. (j) Ours.

and SpecPatConvnet (SPCN) [49]. To demonstrate advantages over other semi-supervised methods, we compare the proposed method with Laplacian SVM (Lap-SVM) [50], DCPN [30], 3DAES [31], and XGPN [34]. We also compare our method with other neural architecture search-based HSI classification methods, such as Auto-CNN [43]. For all comparison methods, we conduct experiments with published source codes or open-source algorithm toolkits. The original hyperparameters of all comparison methods are retained during experiments.

Three evaluation criteria are adopted in this study, including overall accuracy (OA), average accuracy (AA), and Kappa coefficients (κ).

C. Performance Evaluation

In this section, we compare the proposed method with other HSI classification methods in the case of different training samples. The classification performance of different methods on the Indian Pines, University of Pavia, and Houston 2013 is summarized in Tables III–VIII. At the same time, in order to make an intuitive comparison, we visualize the classification results of different methods in Figs. 5–10, where the subscripts of the pictures represent the results from: ground truth, 3DCNN, PRN, SPCN, LapSVM, DCPN, 3DAES, XGPN, AutoCNN, and Ours.

1) Compared With Other Deep Learning-Based Classification Methods: The proposed dynamic deep feature clustering-based semi-supervised method has advantages over these competing methods. Taking the evaluating indicator OA as an example, compared with the SPCN method, the proposed method achieves 12.42% and 27.12% improvement, respectively, when the sample number of the training set is 5 and 10 on the Indian Pines dataset. On the University of Pavia dataset, the proposed method also achieves 14.08% and 19.39% performance improvement compared with the PRN method. These results prove that the proposed method has advantages over other deep learning methods.

2) Compared With Other Semi-Supervised Methods: In this section, we compared the proposed method with some

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TABLE III

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON INDIAN PINES DATASET WITH FIVE LABELED SAMPLES PER CLASS IN TRAINING SET AND THREE LABELED SAMPLES PER CLASS IN VALIDATION SET

Class	**3DCNN** [47]	**PRN** [48]	**SPCN** [49]	**Lap-SVM** [50]	**DCPN** [30]	**3DAES** [31]	**XGPN** [34]	**Auto-CNN** [43]	**Ours**
1	33.05	45.89	54.67	37.89	38.53	44.57	61.00	44.58	56.86
2	52.17	56.63	61.93	54.82	65.30	50.01	35.75	77.17	78.04
3	57.29	59.17	60.00	77.50	77.50	55.31	72.10	79.45	73.92
4	38.63	46.03	60.41	39.86	47.12	79.06	91.15	75.35	86.09
5	64.00	86.11	84.42	69.68	76.63	93.78	93.47	99.27	98.19
6	48.45	71.55	66.70	47.73	55.67	54.46	46.06	74.34	86.42
7	39.59	56.62	53.73	34.34	41.51	52.70	71.26	74.93	65.28
8	56.27	59.32	69.15	61.19	71.86	60.78	28.21	84.49	85.61
9	52.33	73.99	69.49	65.45	64.74	92.83	93.75	79.11	92.42
OA	44.56	59.90	61.56	48.63	53.94	61.73	62.97	70.71	**74.13**
AA	49.09	61.70	64.50	54.27	59.87	64.83	65.86	76.52	**80.32**
κ	38.09	53.97	55.99	41.77	47.61	55.73	57.67	64.92	**69.27**

TABLE IV

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE INDIAN PINES DATASET WITH TEN LABELED SAMPLES PER CLASS IN TRAINING SET AND FIVE LABELED SAMPLES PER CLASS IN VALIDATION SET

Class	**3DCNN** [47]	**PRN** [48]	**SPCN** [49]	**Lap-SVM** [50]	**DCPN** [30]	**3DAES** [31]	**XGPN** [34]	**Auto-CNN** [43]	**Ours**
1	33.82	45.54	37.61	57.82	41.96	56.49	73.42	66.23	74.51
2	61.93	60.00	68.92	75.66	69.52	66.07	58.23	83.84	86.10
3	65.00	63.33	88.12	72.71	76.88	74.50	87.93	96.86	99.52
4	45.34	53.01	47.67	66.71	51.78	90.57	95.66	92.98	96.71
5	72.42	94.23	80.21	97.63	84.42	95.62	96.84	99.98	99.78
6	56.49	74.95	56.60	77.84	62.58	59.62	64.91	84.70	90.20
7	35.03	64.77	43.71	66.88	49.16	59.63	79.26	80.43	76.90
8	68.14	61.53	78.31	80.00	79.49	72.81	42.82	87.77	93.38
9	61.34	79.45	77.31	83.24	73.20	90.76	99.06	92.08	96.41
OA	49.56	64.76	57.72	72.41	60.00	70.11	75.00	82.36	**84.84**
AA	55.50	69.31	64.27	75.39	65.44	72.94	77.57	87.21	**90.39**
κ	42.78	59.34	51.88	68.17	54.41	64.80	71.10	78.67	**81.72**

TABLE V

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE UNIVERSITY OF PAVIA DATASET WITH FIVE LABELED SAMPLES PER CLASS IN TRAINING SET AND THREE LABELED SAMPLES PER CLASS IN VALIDATION SET

Class	**3DCNN** [47]	**PRN** [48]	**SPCN** [49]	**Lap-SVM** [50]	**DCPN** [30]	**3DAES** [31]	**XGPN** [34]	**Auto-CNN** [43]	**Ours**
1	62.28	69.98	61.27	45.11	63.20	49.90	90.20	60.32	73.80
2	59.53	65.97	43.76	52.23	62.60	86.83	89.91	74.34	73.34
3	65.16	68.02	23.63	53.37	67.68	51.47	42.13	84.26	86.5
4	74.25	67.75	78.27	51.99	72.75	89.31	63.41	59.24	68.56
5	76.65	84.31	87.06	64.83	75.99	99.51	97.93	91.40	87.73
6	66.23	75.26	64.98	56.22	71.06	56.77	44.18	82.68	92.32
7	73.61	72.93	81.20	59.47	76.09	54.14	38.82	89.67	96.19
8	67.69	64.10	69.97	52.83	68.75	67.88	62.32	89.77	87.92
9	68.15	81.80	83.81	59.79	71.64	88.15	98.00	70.06	69.08
OA	63.94	68.90	56.12	53.48	66.23	73.79	66.29	75.20	**79.03**
AA	68.17	72.24	65.99	55.09	69.97	71.55	69.66	77.97	**81.71**
κ	55.85	61.52	47.94	42.62	58.55	65.87	58.35	69.43	**74.34**

semi-supervised methods, such as Lap-SVM [50], DCPN [30], 3DAES [31], and XGPN [34]. According to the experimental results on three datasets, the proposed method maintains advantages over semi-supervised-based methods in most cases. Specifically, compared with XGPN, the OA increased by 11.16% and 9.84% on the Indian Pines dataset and 5.24% and 3.36% on the University of Pavia dataset when the number of labeled samples number is 5 and 10, respectively. These results demonstrate the advantageous performance of the proposed method, which can automatically design network structure and find the optimal network structure, especially in the case of a small number of samples.

3) Compared With Other Neural Architecture Search-Based Methods: In this section, we compare the proposed method with Auto-CNN [43], and the experimental results show that the proposed method achieves good performance on all datasets. Similar to Section IV-C1, take the criteria OA as an example, and the proposed method achieves 3.41% and

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TABLE VI

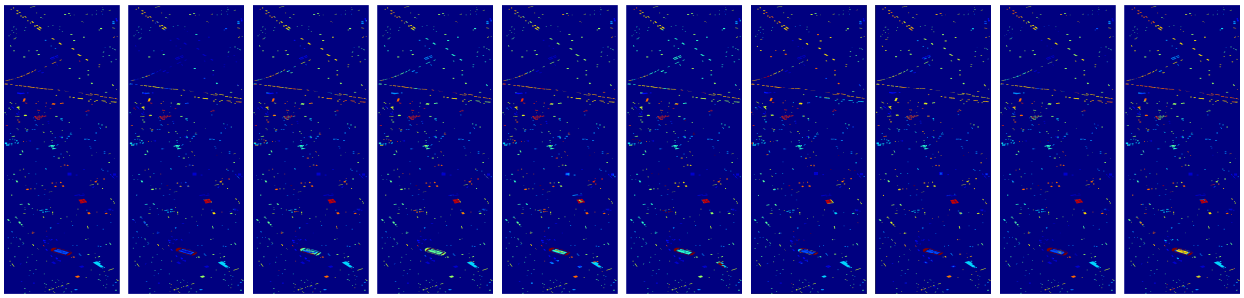
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE UNIVERSITY OF PAVIA DATASET WITH TEN LABELED SAMPLES PER CLASS IN TRAINING SET AND FIVE LABELED SAMPLES PER CLASS IN VALIDATION SET

Class	**3DCNN** [47]	**PRN** [48]	**SPCN** [49]	**Lap-SVM** [50]	**DCPN** [30]	**3DAES** [31]	**XGPN** [34]	**Auto-CNN** [43]	**Ours**
1	75.99	82.87	70.32	62.73	71.22	68.97	96.31	78.47	88.8
2	66.01	73.27	53.66	72.06	66.62	85.78	93.71	79.42	82.24
3	74.75	73.03	34.61	78.95	75.66	46.84	68.30	83.51	91.50
4	83.56	78.50	97.42	75.52	79.54	93.34	81.56	74.26	79.85
5	86.32	89.07	98.94	89.44	86.99	99.31	99.75	94.29	93.56
6	77.35	83.20	76.56	80.94	79.2	76.55	67.07	89.24	98.57
7	75.86	86.84	97.21	72.03	80.53	68.82	68.32	95.51	98.58
8	76.41	72.93	82.69	79.24	75.79	79.80	77.25	94.39	92.68
9	80.42	88.89	97.55	81.59	75.03	95.10	99.79	83.53	83.10
OA	72.74	76.52	67.40	73.62	72.23	80.31	84.54	83.21	**87.90**
AA	77.41	80.96	78.77	76.94	76.73	79.39	83.53	85.85	**89.88**
κ	66.28	71.78	60.71	66.99	65.58	74.49	80.00	79.17	**85.04**

TABLE VII

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON HOUSTON DATASET WITH FIVE LABELED SAMPLES PER CLASS IN TRAINING SET AND THREE LABELED SAMPLES PER CLASS IN VALIDATION SET

Class	**3DCNN** [47]	**PRN** [48]	**SPCN** [49]	**Lap-SVM** [50]	**DCPN** [30]	**3DAES** [31]	**XGPN** [34]	**Auto-CNN** [43]	**Ours**
1	67.94	58.40	72.69	68.34	61.39	74.00	69.18	72.44	69.20
2	64.72	62.42	71.00	71.69	59.85	60.37	86.24	47.78	67.10
3	70.65	58.38	85.90	82.65	68.91	100.00	96.19	92.02	92.74
4	69.26	62.22	78.67	78.84	65.83	85.79	77.83	45.00	57.71
5	72.73	72.14	78.50	79.85	70.32	92.24	84.70	97.31	96.93
6	75.10	60.98	85.17	87.13	71.19	79.38	89.51	75.07	75.85
7	56.79	60.24	64.46	59.78	59.38	76.48	70.76	40.41	54.28
8	38.08	43.57	22.77	37.64	47.07	19.37	65.94	25.48	33.42
9	62.15	47.91	66.55	64.42	59.21	87.81	36.03	64.20	63.35
10	28.82	46.91	41.89	35.97	30.93	45.99	37.37	55.09	63.19
11	50.19	39.41	55.86	56.76	48.48	51.14	29.87	58.13	76.25
12	48.99	62.25	40.33	61.36	44.61	39.25	33.35	44.56	60.13
13	58.32	56.49	40.84	61.05	66.46	91.38	59.51	72.69	73.77
14	77.57	57.49	82.83	83.32	73.28	66.67	93.71	100.00	100.00
15	72.56	53.95	83.09	83.21	71.92	61.22	87.87	86.06	81.89
OA	58.00	55.75	61.72	64.07	56.87	66.25	62.55	59.17	**66.95**
AA	60.93	56.18	64.70	67.47	59.92	68.74	67.87	65.08	**71.05**
κ	54.69	52.25	58.71	61.11	53.50	63.55	59.54	55.95	**64.31**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Classification maps of different methods on the Houston 2013 dataset with five labeled samples per class in the training set and three labeled samples per class in the validation set. (a) Ground truth. (b) 3DCNN. (c) PRN. (d) SPCN. (e) LapSVM. (f) DCPN. (g) 3DAES. (h) XGPN. (i) AutoCNN. (j) Ours.

2.48% performance improvements on the Indian pine dataset, 3.83% and 4.69% on the University of Pavia dataset, and 7.78% and 9.62% on the Houston 2013 dataset with five and ten labeled training samples, respectively. Experimental results show that the proposed method can automatically design the optimal network structure, and especially in the case of small samples, the proposed method performs better by combining semi-supervised learning.

D. Performance With Different Numbers of Labeled Samples per Class for Training

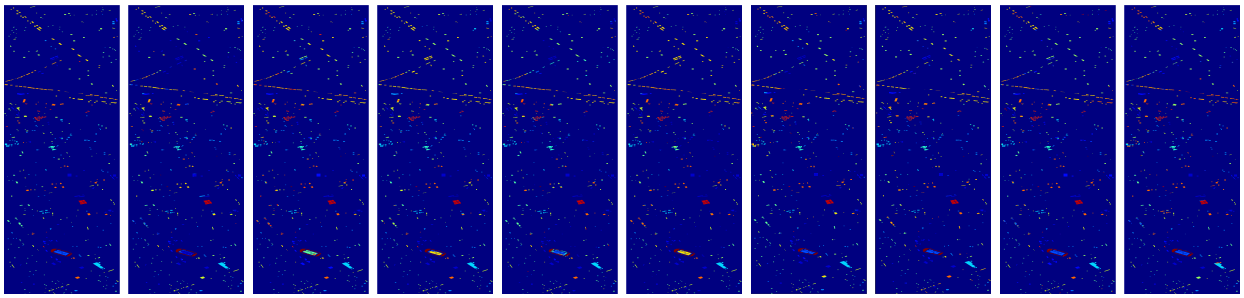
In this section, we discuss to explore the classification performance of the proposed method with different numbers of labeled samples per class for training. Following the previous study [51], we conduct the experiments with five, ten, 15, 20, 25, and 30 labeled samples per class, and the results are shown in Table IX. It can be seen that the classification performance

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TABLE VIII

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET WITH TEN LABELED SAMPLES PER CLASS IN TRAINING SET AND FIVE LABELED SAMPLES PER CLASS IN VALIDATION SET

Class	**3DCNN** [47]	**PRN** [48]	**SPCN** [49]	**Lap-SVM** [50]	**DCPN** [30]	**3DAES** [31]	**XGPN** [34]	**Auto-CNN** [43]	**Ours**
1	80.53	76.49	77.91	79.77	72.00	80.77	87.18	68.12	76.99
2	79.14	77.12	77.05	81.39	72.88	66.71	89.38	65.82	74.07
3	86.22	74.02	91.84	89.50	83.92	97.51	98.94	95.44	93.51
4	85.83	82.90	83.58	97.44	83.18	86.03	95.12	55.10	68.79
5	88.05	91.12	93.24	97.06	88.50	89.85	90.52	98.23	97.83
6	88.63	78.74	95.24	98.60	89.93	89.46	90.02	78.31	79.86
7	71.21	73.15	66.92	65.21	71.36	80.06	78.06	60.72	72.11
8	43.13	57.11	41.14	46.82	61.92	54.76	75.73	33.59	49.14
9	73.84	55.22	67.31	66.38	72.31	79.92	47.29	67.65	76.26
10	39.48	53.96	45.85	43.44	35.91	63.90	52.04	57.82	63.42
11	65.60	51.06	64.02	65.46	56.74	60.69	44.92	59.52	86.88
12	61.94	79.85	68.38	66.76	55.60	73.65	43.66	74.48	85.14
13	78.67	69.82	65.89	66.67	76.21	91.90	78.15	80.77	80.00
14	91.42	65.02	89.47	85.02	89.39	86.73	95.09	100.00	100.00
15	90.61	59.75	89.64	92.39	83.26	83.97	87.50	81.36	89.11
OA	70.53	69.62	70.44	69.23	69.38	75.91	73.19	67.30	**76.92**
AA	74.95	69.69	74.50	73.25	72.87	78.44	76.91	71.80	**79.54**
κ	69.13	67.15	68.73	69.58	66.94	74.98	71.03	64.67	**75.03**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. Classification maps of different methods on the Houston 2013 dataset with ten labeled samples per class in the training set and five labeled samples per class in the validation set. (a) Ground truth. (b) 3DCNN. (c) PRN. (d) SPCN. (e) LapSVM. (f) DCPN. (g) 3DAES. (h) XGPN. (i) AutoCNN. (j) Ours.

will increase as the number of labeled samples increases. It is noticeable that in the case of fewer label samples, the performance improvement of the proposed method is more significant. The reason can be attributed as follows. When only few labeled samples are provided, the unlabeled samples are much more important to help the semi-supervised method to improve the HSI classification performance. However, with more labeled sample introduced, labeled samples begin to dominate the classification performance. As a result, the performance improvement will decrease when more labeled samples introduced.

TABLE IX

CLASSIFICATION RESULTS ON THE INDIAN PINES DATASET WITH DIFFERENT NUMBERS OF LABELED SAMPLES PER CLASS IN TRAINING SET

Dataset	Number of label samples	**5**	**10**	**15**	**20**	**25**	**30**
IndianP	OA	74.13	84.84	89.69	91.67	92.44	93.56
	AA	80.32	90.39	94.00	95.43	96.29	96.59
	κ	69.27	81.72	87.47	89.92	90.75	92.10
PaviaU	OA	79.03	87.90	93.70	95.63	96.81	97.58
	AA	81.71	89.88	95.34	95.63	97.72	98.04
	κ	74.34	85.04	92.11	94.48	95.99	96.95
Houston	OA	66.95	76.92	78.08	79.23	80.52	80.66
	AA	71.05	79.52	80.86	82.20	83.27	83.56
	κ	64.31	75.03	76.32	77.57	78.97	79.15

E. Ablation Study

To further validate the proposed method, a series of ablation experiments is performed in this section. The experiments are performed on the Indian Pines dataset with ten labeled samples per class and the results are the average results on ten rounds of experiments.

1) Influence of Dynamic Feature Clustering and Neural Architecture Search: To illustrate the effectiveness of the two key modules of the proposed method, including dynamic feature clustering and neural architecture search, we conduct ablation experiments on these two modules. The experimental results are shown in Table X, where the second to fourth columns, respectively, represent the proposed method, a method that directly utilizes the original pixel instead of deep feature for clustering, a dynamic feature clustering method-based handcrafted network with a depth same to our network depth, and an original pixel clustering-based handcrafted network. It can be seen that when dynamic feature clustering and neural structure search are employed separately, HSI classification performance improves, and the increase is

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TABLE X
INFLUENCE OF DYNAMIC FEATURE CLUSTERING AND NEURAL ARCHITECTURE SEARCH

Deep feature clustering	✓	✓	✗	✗
NAS	✓	✗	✓	✗
OA	**84.84**	82.46	84.23	82.03
AA	**90.39**	88.85	89.82	88.28
κ	**81.72**	78.80	80.27	77.73

TABLE XI
INFLUENCE OF TWO POSTPROCESSING OPERATIONS

Loss-reweighting	✓	✓	✗	✗
Reclustering	✓	✗	✓	✗
OA	**84.84**	84.57	84.70	84.11
AA	**90.39**	89.91	90.31	89.58
κ	**81.72**	81.32	81.49	80.92

TABLE XII
INFLUENCE OF THE NUMBER OF CLUSTERS IN CHARACTERISTIC CLUSTERING

Cluster Number	**10**	**20**	**50**	**100**
OA	84.56	**84.84**	83.98	82.67
AA	90.30	**90.39**	89.85	89.68
κ	81.59	**81.72**	80.55	78.93

even larger when both are used together. The above experimental results demonstrate the effectiveness of the proposed dynamic feature clustering and neural structure search.

2) Influence of Loss Reweighting and Reclustering: In Section III-B, we have introduced two postprocessing mechanisms after feature clustering, including loss reweighting and reclustering. In order to explore the effectiveness of these two mechanisms, relevant experiments are carried out in this section, and the experimental results are shown in Table XI. The experimental results demonstrate that the network performance is improved to a certain extent when the two mechanisms are applied separately, and the network performs better when these two mechanisms are applied simultaneously. Therefore, we employ both loss-reweighting and reclustering mechanisms in our experiments.

3) Influence of Cluster Number: In order to verify the influence of cluster number on the network classification performance, different numbers are tested in this section. As shown in Table XII, the number of clustering categories has a great influence on the classification ability. Specifically, as the number of clusters increases, the performance of the HSI classification network initially improves, then declines, and reaches the best when the number of clusters is 20. It can be inferred that if the number of clusters is too small, samples belonging to different categories will be clustered together. If the number of clusters is too large, samples of the same category will be separated. Therefore, we set the number of clusters to 20 in our experiments in order to obtain better performance.

4) Influence of Clustering Weights for the Clustering Loss: The loss function of the method proposed in this article consists of two parts: supervision classification loss (L_{cls}) and clustering loss (L_{clu}), as shown in (1). In order to verify the effectiveness of clustering loss on classification results, we conduct experiments with different weights λ in this section. As shown in Table XIII, the performance of HSI classification network varies with the weight λ. According to the result, we set the weight λ to 1 for all experiments in this study to ensure the HSI classification performance.

TABLE XIII
INFLUENCE OF DIFFERENT LOSS WEIGHTS

λ	**0.25**	**0.5**	**1**	**2**	**4**
OA	84.27	84.06	**84.84**	84.12	83.88
AA	90.12	90.28	**90.39**	90.32	89.81
κ	80.93	80.57	**81.72**	80.62	79.31

TABLE XIV
INFLUENCE OF UPDATE WEIGHT OF THE FEATURE MEMORY MODULE

β	**0**	**0.3**	**0.5**	**0.7**	**1**
OA	84.62	**84.84**	83.78	84.15	82.92
AA	90.26	**90.39**	89.31	90.12	88.92
κ	81.48	**81.72**	80.50	80.96	79.39

TABLE XV
INFERENCE TIME OF ALL METHODS ON INDIAN PINES DATASET WITH 20 SAMPLES PER CLASS TO INFERENCE

Methods	3DCNN	PRN	SPCN	LapSVM	DCPN	3DAES	XGPN	AutoCNN	Ours
Times(s)	1.8415	2.9527	0.3721	1.0524	1.9635	0.6054	0.8770	0.8415	0.9599

5) Influence of Update Weight of the Feature Memory Module: During the training process, the features in the feature memory module are constantly changed, which may result in unstable training. To alleviate this issue, we implement soft updates through parameters, as shown in (2). In order to explore the influence of parameter β on the proposed method, we evaluate the proposed method in several rounds of experiments with different β on the Indiana Pines dataset. As shown in Table XIV, the proposed method yields the best performance when $\beta = 0.3$. As a result, we fixed $\beta = 0.3$ for simplicity in all experiments.

F. Running Time

In this section, we compare the running time of inference with different methods on the Indian Pines dataset with 20 samples per class. All experiments were run on a platform with Intel Xeon E5-2650 CPU and GeForce GTX 1080ti GPU. It is worth pointing out that LapSVM uses the CPU only, whereas other deep learning methods use the GPU for inference. As shown in Table XV, the running time of the proposed method is moderate compared with other deep learning-based methods.

V. CONCLUSION

In this study, we propose a semi-supervised structure-adaptive HSI classification method, which can automatically construct a network structure better suitable for the given HSI. Specifically, we propose a two-head semi-supervised HSI classification framework utilizing both

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labeled and unlabeled data, consisting of a shared feature extraction module, a classifier module for labeled samples together with a clustering module for unlabeled samples. To boost the performance of the constructed two-head network, we propose to utilize deep features instead of the original pixels for HSI clustering to generate pseudo-labels for the unlabeled data. Within the conducted semi-supervised network, we specifically design a method to automatically search for the shared feature extraction module better suitable for the given HSI data, which leads to better HSI classification results. Experimental results on three public datasets demonstrate the effectiveness of the proposed method.

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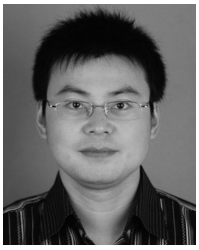
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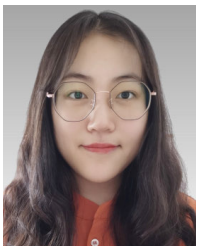
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Wei Wei (Senior Member, IEEE) received the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2012.

He is currently a Professor with the School of Computer Science and Engineering, Northwestern Polytechnical University. He has authored more than 40 papers, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Computer Vision*, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), IEEE International Conference on Computer Vision (ICCV), AAAI Conference on Artificial Intelligence (AAAI), and International Joint Conference on Artificial Intelligence (IJCAI). His research interests include image processing, machine learning, and pattern recognition.

Dr. Wei is a Reviewer of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE REMOTE SENSING LETTERS, and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS. He is also an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Shuyi Zhao received the B.E. degree from Northwestern Polytechnical University, Xi'an, China, in 2021, where she is currently pursuing the M.E. degree.



Songzheng Xu received the M.E. degree from Northwestern Polytechnical University, Xi'an, China, in 2022.

His main research directions include hyperspectral image classification.



Lei Zhang (Member, IEEE) received the Ph.D. degree in computer science and technology from Northwestern Polytechnical University, Xi'an, China, in 2018.

He was a Research Staff with the School of Computer Science, The University of Adelaide, Adelaide, SA, Australia, from 2017 to 2019. He was a Research Scientist with the Inception Institute of Artificial Intelligence (IIAI), Abu Dhabi, United Arab Emirates, from 2019 to 2020. He is currently a Professor with the School of Computer Science, Northwestern Polytechnical University. His research interests include image processing, machine learning, and video analysis.



Yanning Zhang (Senior Member, IEEE) received the B.S. degree from the Dalian University of Technology, Dalian, China, in 1988, the M.S. degree from the School of Electronic Engineering, Northwestern Polytechnical University, Xi'an, China, in 1993, and the Ph.D. degree from the School of Marine Engineering, Northwestern Polytechnical University, in 1996.

She is currently a Professor with the School of Computer Science, Northwestern Polytechnical University. She is also a Cheung Kong Professor of Ministry of Education, China. She has authored more than 200 articles. Her research interests include remote sensing image analysis, computer vision, and pattern recognition.

Dr. Zhang is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE TRANSACTIONS ON MULTIMEDIA AND PATTERN RECOGNITION.

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